

SUSITRA GNANASAMBHANDAM

Healthcare Informatics & Implementation Professional

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M.S. Biomedical Data Science & Informatics (Clemson) · Clinical AI Research, Brigham and Women's Hospital / Harvard Medical School · FHIR R4 · EHR

TECHNICAL SKILLS

EHR & Interoperability: FHIR R4, HL7, ICD-10, CPT, EHR Systems, HIPAA, REDCap, SMART on FHIR, Clinical Decision Support, Health IT
Data & ETL: SQL Server, T-SQL, SSIS, SSRS, Star-Schema Modeling, Data Warehousing, MongoDB, ETL Pipelines, Query Optimization
Analytics & BI: Power BI (DAX, RLS, Power BI Service), Tableau, Excel Dashboards, Statistical Modeling, SAS, R
Programming & Tools: Python (Pandas, NumPy, FastAPI, PyMongo, scikit-learn), REST APIs, Power Automate, Git, Jira, SQLite
Healthcare Domain: Clinical Workflows, Population Health, Value-Based Care, Clinical Trials Methodology, GCP, HIPAA Compliance

EXPERIENCE

Healthcare Analytics Analyst • Cozy Systems

Apr 2025 – Dec 2025

Healthcare Analytics · Clinical & Operational BI · Stakeholder Delivery · SQL Server · SSIS · Power BI | Cumming, GA (Remote)

- Delivered Power BI analytics solutions for healthcare operations clients — partnering with clinical and business stakeholders to translate reporting requirements into validated dashboards used by executive and department leadership.
- Deployed SSIS ETL pipelines integrating SQL Server, Excel, and REST APIs with error handling, audit logging, and retry logic; reduced manual healthcare reporting effort by 40% and ensured audit-ready, compliance-ready data delivery.
- Developed DAX measures, time-intelligence KPIs, and forecasting dashboards for clinical operations leadership; implemented Row-Level Security governance coordinating with healthcare teams on role-based data access controls.
- Optimized T-SQL queries via execution plan analysis and index tuning, reducing report load time ~30%; performed data validation and QA across healthcare reporting workloads.

Analytics Infrastructure Analyst • Cozy Systems

Nov 2023 – Aug 2024

Data Warehouse Architecture · T-SQL · SSIS · Star-Schema Modeling · Semantic Modeling | Chennai, India (Remote)

- Designed a centralized SQL Server data warehouse with star-schema modeling, consolidating multiple source systems into analytics infrastructure supporting enterprise self-service reporting.
- Built T-SQL stored procedures and SSIS packages with structured logging and validation; developed Power BI semantic models with DAX measures enabling scalable reporting across business units.

Biomedical Research Intern • Brigham and Women's Hospital, Harvard Medical School

Apr – Oct 2023

Clinical AI · Point-of-Care Diagnostics · HIPAA-Compliant Dataset Validation · Multidisciplinary Research | Boston, MA

- Validated CNN-based clinical AI model performance within a point-of-care diagnostics pipeline — evaluated model outputs against clinical ground truth, identified data quality gaps, and documented findings for clinical and engineering stakeholders.
- Maintained HIPAA-compliant clinical imaging datasets ensuring traceability and reproducibility; prepared analytical summaries supporting stakeholder presentations and publication workflows.

PROJECTS

FHIR Bulk Data Ingestion Pipeline • Python · MongoDB · FHIR R4 · NDJSON · PyMongo · Pandas

2025

- Built a production-style FHIR R4 bulk ingestion pipeline processing NDJSON exports across 10 clinical resource types — Patient, Observation, Condition, Encounter, MedicationRequest, and 5 additional — into MongoDB, preserving schema integrity for clinical analytics, CDS services, and EHR integration use cases.

Agentic AI System — 26-Tool Finance Agent • Python · FastAPI · Claude API · SQLite · Twilio · Chart.js

Feb 2026

- Architected a 26-tool AI agent handling end-to-end workflows: natural language intake, receipt image analysis via vision API, multi-turn conversation management, and proactive alerting — with a dedicated non-technical end-user interface via WhatsApp/Twilio and a real-time analytics dashboard. Resolved production failures including tool sequencing errors and prompt drift.

Hypertension RCT — Clinical Study Design & Statistical Analysis Plan • Study Design · RCT Methodology · Statistical Analysis · SAS · Prisma Health

2025

- Designed a two-arm randomized controlled trial evaluating lifestyle intervention on diastolic BP in adults 60+; developed full statistical analysis plan (t-tests, correlation, mediation analysis) and expanded into a Prisma Health-aligned seed grant proposal with budget and dissemination strategy.

PUBLICATIONS

Comparing Remote Smartphone-Based Semen Assessments with Laboratory Evaluations

Jul 2025

Mitsunami M., Kandula H., Sivakumar A.D., Gnanasambhandam S., et al. · *Fertility and Sterility* · Brigham and Women's Hospital, Harvard Medical School · Peer-reviewed

EDUCATION

M.S. Biomedical Data Science & Informatics • Clemson University

GPA 3.75 · Dec 2025

Clinical Decision Support Systems · EHR Systems · Health Informatics · Biomedical AI · Clinical Trials Methodology

B.S. Biomedical Sciences (Hons.) • Sri Ramachandra Institute of Higher Education and Research

GPA 8.61/10 · Feb 2024

Molecular Biology · Biomedical Engineering · Physiology · Biochemistry · Biostatistics · Research Methodology

CERTIFICATIONS

GCP for Clinical Trials — FDA Focus • CITI Program • Credential #76844792

Issued May 2026 · Valid through May 2029

HIPAA Basics for Healthcare Professionals • CITI Program • Credential #76844793

Issued May 2026 · Valid through May 2027